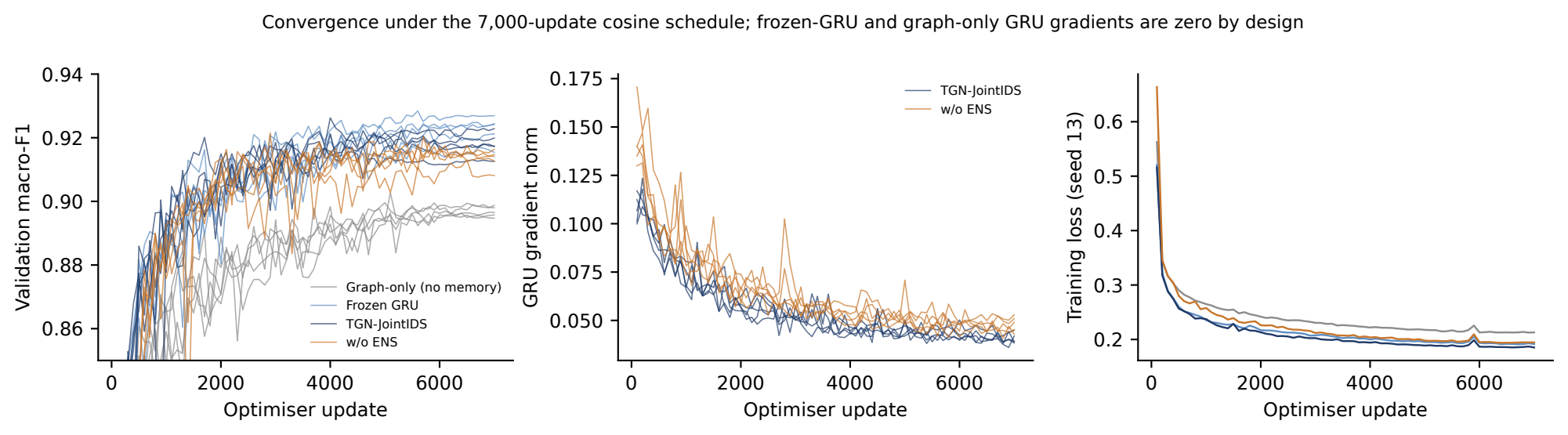
Convergence under the 7,000-update cosine schedule; frozen-GRU and graph-only GRU gradients are zero by design
Validation macro-F1
0.94
0.92
0.90
0.88
0.86
0
2000
4000
6000
Optimiser update
Graph-only (no memory)
Frozen GRU
TGN-JointIDS
w/o ENS
GRU gradient norm
0.175
0.150
0.125
0.100
0.075
0.050
0
2000
4000
6000
Optimiser update
TGN-JointIDS
w/o ENS
Training loss (seed 13)
0.6
0.5
0.4
0.3
0.2
0
2000
4000
6000
Optimiser update